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Top Physics honour for Prof Ashoke Sen, working to unify the laws of universe

Amitabh Sinha
New Delhi, October 24

PROFESSOR ASHOKE Sen is not a household name. But in the small community of theoretical physicists, particularly amongst string theorists, he has a formidable reputation.

As one of the world's leading theoretical physicists working in the area of string theory, the 70-year-old Sen has won several of the top prizes in Physics in the last three decades. The Oskar Klein medal that he has just been honoured with only reinforces his credentials as one of the most prominent scientists in his field.

Now a professor at the Bengaluru-based International Centre for Theoretical Sciences, Sen is credited with some very important breakthroughs in string theory, a still-developing field that is seeking to come up with a new unified theory of the universe that can bridge the inconsistencies between two of the existing grand theories of nature — the theory of gravitation that explains nature at the scale of stars and galaxies, and quantum theory that

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Prof Ashoke Sen, 70, is one of the world's leading theoretical physicists working in the area of string theory

Top Physics honour

applies to the sub-atomic world of fundamental particles.

The two theories, both a century old, have been extremely successful within their own realms, but don't work smoothly when applied together in certain situations. For more than half a century now, scientists have been working to reconcile these inconsistencies, looking for a unified theory that incorporates gravitation as well as quantum theory. String theory is one of the leading efforts in that direction. It explains the fundamental forces and particles as the different vibrations for very small string-like particles, which it describes as the most basic building blocks of the universe.

String theory has been able to achieve some significant progress in creating the theoretical and mathematical framework for bridging the inconsistencies, but unfortunately has not yet been able to come up with any predictions that can be tested and verified through experiments.

Sen, who has a B.Sc in Physics from Calcutta University in 1975 and an M.Sc from IIT Kanpur in 1978, has been working on string theory ever since his return to India in the early 1990s after his PhD and post-doctoral work abroad.

He started working at the Tata Institute of Fundamental Research which already had a strong group of physicists working on string theory. A few years later, he moved to Harish Chandra Institute in Prayagraj, an autonomous institution under the Department of Atomic Energy that was previously called Mehta Research Institute.

Sen attracted global attention, soon after returning to India, when he began publishing some important breakthroughs in the understanding of string theory. String theory itself is no single approach. In fact, there are at least five different approaches to string theory that are theoretically and mathematically consistent. This multiplicity of string theories was seen as a conceptual weakness, because nature could not be expressing itself through so many different kinds of theories.

In the 1990s, Sen was able to show that certain kinds of differences in these theories were actually equivalent, opening up the possibility that these different string theories were essentially multiple descriptions of the same underlying physics. This was a profound development that led to significant progress in the field.

International honours followed almost immediately. At the age of just 42, Sen became a Fellow of the Royal Society (FRS), and six years later Fellow of the Third World Academy of Sciences. He had already been honoured with India's highest scientific award, Shanti Swarup Bhatnagar Prize, in 1994.

His productive years continued in the 2000s, as he made several other contributions to string theory, black hole physics, and particle physics. These contributions got him the Breakthrough Prize in Fundamental Physics in 2014 and the Dirac Medal two years later, two of the topmost prizes in Physics. He was the first Indian recipient of both. In the meanwhile, he had also been honoured with the Infosys Prize in 2009, and Padma Bhushan in 2013.

India has had a fairly strong tradition in theoretical physics, with several groups across institutions working at the frontiers of research. In one of his interviews, Sen pointed out that one of the reasons for this could be that theoretical physics does not require any major equipment or infrastructure.

In 2021, Sen moved from the Harish Chandra Institute to the International Centre for Theoretical Physics in Bengaluru, an institution affiliated with TIFR, where he has continued to develop his work.

"For me, the best part of doing science is that I can work on whatever interests me the most," Sen was quoted as saying, after winning the Breakthrough Prize in 2012.

Digital push: 11 crore farmers get unique IDs

Harikishan Sharma
New Delhi, October 24

THE UNION Ministry of Agriculture and Farmers' Welfare said on Saturday that it has reached its target of creating 11 crore Farmer IDs five months ahead of its scheduled deadline of March 31, 2027.

"In a major milestone for digital agriculture, the Government of India has achieved its target of creating 11 crore Farmer IDs, five months ahead of the March 31, 2027 deadline," the ministry said in a statement on Saturday. Agriculture Minister Shivraj Singh Chouhan said the Farmer ID system would significantly reduce the need for farmers to repeatedly visit government offices and banks or submit the same documents for availing different schemes.

The Farmer ID is an Aadhaar-like unique digital identity that is linked dynamically to the State's land records, and has information like demographic, crop sown, ownership details, etc. The database created through Farmer IDs is known as Farmers' Registry. It serves as one of the three core registries under the Agri-Stack component of the Union Government's Digital Agriculture Mission – a digital public infrastructure initiative for the farm sector that was approved by the Union Cabinet on September 2, 2024.

The other two registries under the Agri-Stack are Geo-referenced Village Maps and Crop Sown Registry.

According to ministry sources, 26% of 11 crore registered farmers, who have been provided Farmers' ID are women. Of the total 11 crore farmers, approximately 81% belong to marginal and small operational-holding categories.

The age-wise analysis of the data shows that around 15% of these 11 crore farmers were 70 years of age or older, while 7.5% were below 30 years. "The majority of the farmers lie between 30-70 years of age," a source said.

Discussing further progress of the Farmer ID initiative, the source said that an important emerging priority is inclusion of actual cultivators who may not be recorded as landowners.

"Already 28,680 cultivators, including eligible tenants and sharecroppers, have been registered... as per the State-decided mechanism for identification of cultivators, such as tenants and sharecroppers," the source said.

Explaining various applications of Farmer' IDs, the source said integration of Agri-Stack with AI, weather forecasts, etc. can enable more accurate crop production forecasts and improved food security planning, personalised agricultural advisories, better market intelligence, and faster disaster response.

INDIVIDUALS ARE SUPPOSED TO BE EXEMPT FROM NEW FOREX TRADE REPORTING NORMS

RBI's trade reporting norms worry individual service exporters, MSMEs

Ravi Dutta Mishra & Siddharth Upasani
New Delhi, October 24

WEEKS AFTER the Reserve Bank of India's (RBI) new foreign exchange reporting regulations for export and import of goods and services took effect, individuals exporting services, as well as micro, small and medium enterprises (MSMEs), are struggling to get their payments credited, as banks are asking them to fill out a complex, five-page export declaration form (EDF) as a prerequisite.

According to the RBI, beginning October 1, an exporter of services must furnish a declaration in the EDF, specifying the full export value of the services within 30 days from the end of the month in which the invoice was raised. The catch, however, is that individuals — such as freelancers and consultants — are exempt from this requirement.

The fact that banks are still asking individuals to fill the form points to a patchy implementation of the Foreign Exchange Management (Export and Import of Goods and Services) Regulations — which, to be sure, were issued as far back as January 13, around eight months before the October 1 deadline.

No FAQ by RBI

Deputy Governor Rohit Jain had, on October 7, said that the intent of the new trade regulations is to "simplify the processes" and "promote ease of doing business".

"So, there seems to be some misunderstanding on the reporting obligations, which we will clarify shortly through an FAQ," Jain had added. However, no FAQs have been issued by the central bank so far. And banks, it seems, are unwilling to oblige despite both Jain and Governor Sanjay Malhotra's public assertions that individuals "irrespective of the amount, are not required" to report the information.

This includes things like personal consumption payments, like paying overseas for foreign newspapers or TV channels or providing services abroad as an individual, such as tuition.

"As already clarified by Rohit (Jain), it's important. That's why I am repeating this for the benefit of your viewers," Malhotra had added on October 7. According to information available on the website of the Foreign Exchange Dealers' Association of India, only two banks — HDFC Bank and State Bank of India — out of 90 provide a policy framework for customers to understand the trade regulations that came into effect from October 1.

MSMEs voice concern

MSMEs, meanwhile, have approached the RBI seeking urgent intervention regarding the Export Declaration Form reporting and reconciliation requirements. India SME Forum president Vinod Kumar, in an October 7 letter to the RBI, said that the final EDF framework does not provide a general turnover-based exemption for micro and small service exporters.



Deputy Governor Rohit Jain has said the intent of the new trade rules is to "simplify the processes" and "promote ease of doing business". REUTERS

"This is not merely an additional form. For a micro enterprise or an individual exporter with a lean team and no dedicated FEMA or compliance function, recurring EDF preparation requires invoice-level records, customer and country details, currency and value data, contract information, service classification, reconciliation of receipts, and subsequent follow-up with the Authorised Dealer bank. The compliance cost can therefore become material in relation to the value of the export itself, particularly for high-frequency, low-value digital service transactions," Kumar said.

"We had also proposed simplified, technology-enabled and proportionate compliance, including de minimis treatment and aggregate/self-declaration mechanisms for smaller exporters," he said.

A social media post by content creator Ramesh Srivats on the troubles he faced in receiving payments from abroad — despite the RBI's clarifications on individuals — has been widely shared. Srivats said that EDF filing is riddled with complexities and runs into five sheets. "At least for me, most of the details are constant. One client, one payment. If it's a one-time exercise, fine. I'll grit my teeth and get it done. But no. These details have to be given month after month. Else... no payment will be credited. It would have been simple enough for RBI or HDFC to create an online portal for this. With dropdowns, saved formats, contextual help, etc. to at least make this easy. But that would be too convenient, no? So, they've chosen to dump the entire burden on the taxpayer," he said. Ajay Srivastava, the head of think tank Global Trade Research Initiative, posted on social media that the RBI should come out with more clarity. "For instance if I receive $100 from the US do I need to file an EDF when it is a gift from my brother, payment for tutoring students or a fee for management consultancy," Srivastava said.

India SME Forum also asked RBI to prescribe a common digital procedure, definitive document checklist, transparent and capped charges, time-bound closure standards and an effective escalation mechanism.

> **"For a micro enterprise or an individual exporter with a lean team and no dedicated FEMA or compliance function... The compliance cost can therefore become material in relation to the value of the export itself, particularly for high-frequency, low-value digital service transactions**
>
> **VINOD KUMAR**
> PRESIDENT, INDIA SME FORUM

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